



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations

May 2024

EC1002-2 – APPLIED DIGITAL LOGIC DESIGN
 (ACT, AD, AM, CB, CC, CS, EC, EE, IS, RI, VLSI)

Duration: 3 Hours

Max. Marks: 100

Note:

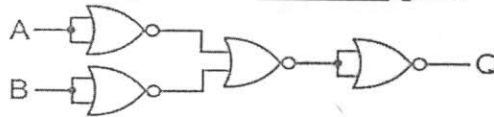
Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Adding 1010 and 0011 gives the output of
 A) 0111
 B) 1011
 C) 1101
 D) 1001
- Binary subtraction of 0-1 yields
 A) Difference = 0, borrow = 0
 B) Difference = 1, borrow = 0
 C) Difference = 1, borrow = 1
 D) Difference = 0, borrow = 1
- If A and B are the inputs of a half adder, the sum is given by ____
 A) A XOR B
 B) A OR B
 C) A XNOR B
 D) A AND B
- The Boolean expression $AB+AC'+BC$ simplifies to
 A) $BC+AC'$
 B) $AB+AC'+B$
 C) $AB+A'$
 D) $AB+BC$
- The output of the given logic circuit represents _____ gate.



- _____ are universal logic gates.
 A) NAND and NOR
 B) NOR AND EX-OR
 C) EX-OR and EX-NOR
 D) NOT and AND
- A 4x1 Mux will have ____ select lines
 A) 4
 B) 2
 C) 3
 D) 1
- Which of the following expressions represent exclusive NOR of x and y?
 A) $(x \text{ xor } y)'$
 B) $x \text{ xor } y$
 C) $x \text{ or } y$
 D) $x \text{ and } y$
- The truth table for an S-R flip-flop has how many VALID entries?
 A) 3
 B) 2
 C) 1
 D) 4
- A 3 variable Karnaugh map has _____
 A) 18 cells
 B) 16 cells
 C) 8 cells
 D) 3 cells
- One way to make a four-bit adder(A+B) constructed using full adders to perform subtraction(A-B) is by:
 A) Inverting the output
 B) Inverting the carry-in
 C) Inverting the B inputs
 D) Without any changes

12. Which is the main objective of multiplexer circuit?
 A) Decoding the binary information B) Generation of all minterms in an output function with OR-gate
 C) Generation of selected path between multiple sources and a single destination D) Encoding of binary information
13. If an OR gate has 4 inputs with one input high and the other three low, then the output is
 A) LOW B) HIGH
 C) HIGH or LOW depending on the input magnitude D) Alternately HIGH and LOW
14. In which one of the following counters, the flip flops are not clocked simultaneously?
 A) Synchronous counter B) Asynchronous counter
 C) Positive edge triggered counter D) Negative edge triggered counter
15. The characteristic equation for T flip flop is
 A) $TQ + T\bar{Q}$ B) $\bar{T}Q + T\bar{Q}$
 C) $\bar{T}Q + T\bar{Q}$ D) $T\bar{Q} + TQ$
16. If a three-variable switching function is expressed as the product of max terms by $f(A,B,C) = \pi(0,3,5,6)$ then it can also be expressed as sum of min terms by _____.
 A) $\pi(1,2,4,7)$ B) $\Sigma(0,3,5,6)$
 C) $\Sigma(1,2,4,7)$ D) $\Sigma(1,2,3,7)$
17. The characteristic equation for D flip flop is
 A) D B) DQ
 C) $\bar{D}$ D) $D\bar{Q}$
18. A circuit that converts n inputs to 2^n outputs is called
 A) Encoder B) Decoder
 C) Comparator D) Adder
19. A serial in parallel out, 4-bit shift register initially contains all 1s. The data nibble 0111 is waiting to enter. After four clock pulses, the register contains _____.
 A) 0111 B) 0101
 C) 1110 D) 1100
20. Number of flip-flops required to design a counter to count from 0 to 2 is
 A) 4 B) 2
 C) 6 D) 8

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Simplify the following using Boolean algebra:
$(A + B' + C')(A + B' + C)(A + B + C')$. | 6 | L*3 | 1 | 1 |
| b) Prove De-Morgan's theorems using the truth table approach. | 6 | L2 | 1 | 1 |
| c) Minimize the given logic function using the K-map method.
$F(A,B,C) = A'BC' + ABC' + ABC$. | 4 | L3 | 1 | 1 |
| 2. a) Given the simplified expression of a Boolean function, write the truth table, and minterm list and obtain the given simplified function first by writing the equation in canonical form and then use K-map to simplify.
$Y = f(a, b, c) = c'$. | 6 | L3 | 2 | 1 |
| b) Use 2's complement method to find: (i) 1101-1001 (ii) 1001-1101. | 6 | L3 | 2 | 1 |
| c) Perform the conversion for the following and show each step clearly.
(i) $1A2_{16} = (?)_{10}$ (ii) $45.25_{10} = (?)_2$ | 4 | L2 | 1 | 1 |
| 3. a) A system receives three inputs and generates a one-bit output based on the even number of ones present in the input. If the inputs are completely zeros or if it has an odd number of ones, the output will be zero and if the input has an even number of ones then the output will be set to one. Write the truth table for such a system and represent the simplified function using the logic diagram. | 6 | L3 | 2 | 1 |

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|--|---|----|---|---|
| b) Simplify the logic function $f(a,b,c) = \sum m(0,2,3,4,6,7)$ using QM technique | 6 | L3 | 2 | 1 |
| c) Use K-map to minimize the logic function $f(a,b,c) = \sum m(0,2,3,4,6,7)$. | 4 | L3 | 2 | 1 |

Unit – II

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|--|---|----|---|---|
| 4. a) Implement the given function using an active high decoder: | | | | |
| i. $F1(a,b,c) = \sum m(1,2,6)$ | | | | |
| ii. $F2(a,b,c) = \sum m(0)$ | | | | |
| iii. $F3(a,b,c) = \sum m(3,4,5)$ | 6 | L3 | 3 | 1 |
| b) Convert: | | | | |
| i) JK flip flop to D Flip Flop, | 6 | L3 | 4 | 1 |
| ii) JK Flip Flop to T Flip Flop. | 4 | L2 | 3 | 1 |
| c) Explain the working of 4 to 2 Encoder | | | | |
| 5. a) Explain with an example the working of 4-bit binary parallel adder/subtractor with relevant block diagram. | 6 | L2 | 3 | 1 |
| b) Explain the working of edge triggered D flip flop with relevant logic diagram and function table. | 6 | L2 | 4 | 1 |
| c) Explain the working of 2x1 Multiplexer. | 4 | L2 | 3 | 1 |
| 6. a) Implement a Full Adder using a 4x1 multiplexer. | 6 | L3 | 3 | 1 |
| b) Explain the operation of master-slave JK flip-flop with truth table and timing diagram. | 6 | L2 | 4 | 1 |
| c) Obtain the characteristic equation of a JK flip flop. | 4 | L3 | 4 | 1 |

Unit – III

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|---|---|----|---|---|
| 7. a) Explain the working of a 3-bit ripple-up counter using the positive edge-triggered T flip-flops with a relevant logic diagram and the count sequence. | 8 | L2 | 5 | 1 |
| b) Explain the operations of the 4-bit serial in parallel out type shift register using D flip flops with relevant circuit diagrams and suitable examples. | 8 | L2 | 5 | 1 |
| 8. a) Design a synchronous 3-bit up counter using positive edge triggered D flip flops. | 8 | L3 | 5 | 1 |
| b) Design a synchronous 3-bit down counter using positive edge triggered T flip flops. | 8 | L3 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
